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IMPLEMENTATION OF PATIENT REPORTED OUTCOME MEASURES AND PATIENT REPORTED EXPERIENCE MEASURES IN MELANOMA CLINICAL QUALITY REGISTRIES: A SYSTEMATIC REVIEW

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IMPLEMENTATION OF PATIENT REPORTED OUTCOME MEASURES AND PATIENT REPORTED EXPERIENCE MEASURES IN MELANOMA CLINICAL QUALITY REGISTRIES: A SYSTEMATIC REVIEW

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ABSTRACT

Objectives: To identify patient-reported outcome measures (PROMs) and patient-reported experience measures (PREMs) in clinical quality registries, for people with cutaneous melanoma, to inform a new Australian Melanoma Clinical Outcomes Registry (MelCOR); and describe opportunities and challenges of routine PROMs/PREMs collection, especially in primary care.

Design: Systematic review.

Primary and secondary outcome measures: Which PROMs and PREMs are utilised in clinical quality registries for people with cutaneous melanoma, how they are collected, frequency of collection, participant recruitment methods and funding models for each registry.

Results: Of 1134 studies, 14 were included after screening, identifying four relevant registries: Dutch Melanoma Registry, Adelphi Real-World Disease-Specific Programme™ (Melanoma), Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) Registry, and Cancer Experience Registry. These utilised seven PROMs: EQ-5D, Functional Assessment of Cancer-General (FACT-G) and -Melanoma (FACT-M), European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-C30 (EORTC QLQ-C30), Fatigue Assessment Scale (FAS), Hospital Anxiety and Depression Scale (HADS), Patient Reported Outcome Measures Information System-29 (PROMIS-29) and one PREM; EORTC Quality of Life Questionnaire – Information Module 26 (EORTC QLQ-INFO26). PROMs/PREMs in registries were reported to improve transparency of care; facilitate clinical auditing for quality assessment; enable cost-effectiveness analyses and

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create large-scale research platforms. Challenges included resource burden for data entry; and potential collection bias toward younger, more affluent respondents.

Conclusions: Clinical registries indicate PROMs/PREMs for melanoma care can be incorporated and address important gaps, however cost and collection bias may limit generalisability.

Trial registration: International Prospective register of systematic reviews registration (PROSPERO) registration number CRD42018086737.

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ARTICLE SUMMARY

STRENGTHS AND LIMITATIONS OF THIS STUDY

- Patient reported outcome measures (PROMs), have been highlighted by melanoma consumers (patients and carers) as important measures of quality that should be incorporated into clinical quality registries
- This study captures publications of PROM and PREM collection across a range of database types.
- Only clinical quality registries which have appeared in published works with melanoma-specific search terms were identifiable and included.

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INTRODUCTON

Patient-reported outcome measures (PROMs) are described as "measurements of any aspect of a patient's health status that come directly from the patient"(1). These tools are commonly used to assess patients' self-perception of their functional status, health or quality of life. Patient-reported experience measures (PREMs) focus on patients' satisfaction with care and cover topics such as dignity and respect; consistency and coordination of care; adequate involvement in, and explanation of care; trust and communication with nurses and doctors; and satisfactory discharge information (2). Whilst these measures were originally developed for use in research, increasingly they are being utilised for broader purposes, such as inclusion in routine primary care or clinical quality registries for quality assurance and benchmarking (3).

PROMs and PREMs can be applied in several stages, such as initial patient evaluations, during treatment, and in follow-up (4). In the case of cancer, the Prostate Cancer Outcomes Registry – Australia and New Zealand (PCOR-ANZ) uses PROMs to assess the impact of diagnosis and treatment in men with prostate cancer towards improving patient wellbeing (5). Qualitative data collected by this registry has resulted in tailored strategies and targeted programs to address important gaps in patient care.

With the support from key partner organisations, funding was raised through the Melanoma March event to develop the Australian Melanoma Clinical Outcomes Registry (MelCOR). Collection of PROMs and PREMs is an important goal of MelCOR, however challenges in this are apparent. PROMs and PREMs are not routinely collected as a part of melanoma care (including primary care) in Australia, which tends to focus mostly on

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traditional outcomes such as prevention of recurrence, diagnosing subsequent primary melanomas (6) and maximizing life expectancy after diagnosis. Surgical morbidity parameters such as wound complications, pain and numbness are often added to these traditional measures (7). However, self-reporting of patient diagnosis and treatment experiences are typically not evaluated, despite this being an important indicator of quality of care. For example, patient's self-perception of keloid scarring following diagnostic excision or other surgery on the face or hands may be associated with significant psychological morbidity including anxiety, poor self-esteem and a decreased quality of life (8).

As the value and usefulness of MelCOR will be enhanced by inclusion of PROMs and PREMs, we sought to understand which tools capture the most relevant generic and disease-specific information for people with cutaneous melanoma, and the opportunities and challenges associated with routine PROMs and PREMs collection. The aim of this systematic review was to identify and evaluate PROMs and PREMs that are used in clinical quality registries and that might be applied to patients with early-stage cutaneous melanoma, including in primary care settings where much of the initial evaluation and management of this cancer occurs.

METHODS

A systematic review of literature was conducted. The search strategy included patients treated for cutaneous melanoma of any stage, along with articles discussing the advantages and challenges of PROMs or PREMs implementation in registries. (Appendix 1) Five databases were searched from inception to December 2017: MEDLINE (including PreMEDLINE), Embase,

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PsychInfo, Cochrane Database, and TUFTS Cost-Effectiveness Analysis Registry. Reference lists of relevant articles were hand searched for additional publications. A secondary search of MEDLINE (including PreMEDLINE), EMBASE, PsychInfo and the Cochrane Database was conducted from December 2017 until February 2020. Guidelines for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (9) were followed where applicable. The review was registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42018086737).

Patient and Public Involvement Statement

Two melanoma patients provided feedback on the importance of this review and its relevance to patient outcomes.

Search strategy

'Patient reported outcome measures' and 'patient reported experience measures' (including abbreviations PROM or PREM), along with related keywords for patients (consumer, participant) and outcomes (satisfaction, experience, assessment, symptom-measure), or quality of life, were combined with Boolean logic (AND/OR) with terms for melanoma (including lentigo-maligna), to identify relevant studies. Medical Subject Headings were adapted to each database as appropriate (Figure 1 – PRISMA Flow chart).

Eligibility criteria

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Publications involving the use of PROMs and/or PREMs in clinical registries reporting adults with cutaneous melanoma of any disease stage (i.e. stage 0-IV), were included. PROMs and PREMs completed by caregivers (proxy) were also included. Mixed cancer populations that included cutaneous melanoma were eligible, however non-cutaneous melanoma populations (e.g. uveal melanoma), and patients with non-melanoma skin cancers only, were excluded. The use of clinician-reported outcomes was excluded, as were PROMs/PREMs collected solely for research purposes (e.g. within a clinical trial). All primary study types were eligible, and grey literature was also searched, with government reports and digital theses included. Cost-effectiveness analyses were included up to the full-text screening stage, and their reference lists hand-searched for relevant articles. Review articles were excluded.

Study screening and data extraction

Citations and abstracts were imported into Covidence (Cochrane systematic review management program) for screening purposes. Title and abstracts were screened by two independent reviewers (ZB, AT), with discrepancies resolved through discussion, and consultation with a third reviewer (RLM) where necessary. Full-text articles were assessed if the paper was deemed to meet the inclusion criteria, was a cost-effectiveness study, or the abstract did not contain sufficient information. For each clinical registry identified through this review, the PROMs/PREMs used, patient demographics, method of collection, collection time points, and purpose of the registry were identified (Table 2). Any descriptive text relating to advantages or challenges of registry collection were also extracted. The consensus-based standards for the selection of health status measurement instruments (COSMIN) four-point checklist and International Society for Quality of Life Research (ISOQOL) user's guide to

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implementing patient-reported outcome assessment in clinical practice were not used to assess instrument quality, as a critique of PROMs/PREMs used in registries was not an aim of this systematic review. Risk of bias in individual studies was described; studies were not excluded due to bias.

RESULTS

Database and hand-searching yielded 1134 articles, with 988 articles remaining after removal of duplicates (Figure 1). Following abstract and full-text screening, 14 relevant articles remained for analysis. These studies used either data collected from these registries to answer a research question, or were articles commenting on the advantages and challenges of PROM/PREM collection and implementation via a registry. A summary is provided in Table 1. Four relevant clinical registries were identified: The Dutch Melanoma Treatment Registry (DMTR), Adelphi Real-World Disease-Specific Programme (Melanoma), the Cancer Experience Registry, and the Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) registry. Across these four registries, ten PROMs and one PREM were used. A summary of findings is presented in Table 2.

The Dutch Melanoma Treatment Registry

The DMTR is a registry of patients based in the Netherlands with unresectable stage IIIc or IV melanoma. It is used for clinical auditing, improving the transparency of quality of care in the treatment of melanoma, to detail the effects and costs associated with treatment,

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and serves as a database for melanoma research. This registry collects three PROMs: EQ-5D, Functional Assessment of Cancer Therapy – General (FACT-G) and Melanoma (FACT-M). It also collects treatment activity through a Resource Use Questionnaire–Melanoma (RUQ-M). The DMTR aims to collect PROMs every three months and the RUQ-M every six months. Questions about employment and time lost from work due to melanoma and its treatment are used to quantify productivity gains and losses, and are often used in economic evaluations undertaken from a societal perspective. Data are prospectively collected using a web-based program from patients diagnosed with metastatic melanoma who attend participating treatment centres.

The Adelphi Real-World Disease Specific Programme (Melanoma)

The Adelphi Real-World Disease-Specific Programme (Melanoma) is a privately owned international healthcare database used for research purposes. Adelphi collates disease specific programmes on over 60 diseases, including melanoma, with data used for both medical and market research. The Melanoma programme collects three PROMS; the EQ-5D, FACT-G and FACT-M from patients with all stages of melanoma under the care of participating clinicians, from a range of countries. Here, participating physicians are asked to collect data on the next 10 patients being consulted for the disease of interest, with these patients asked to self-complete the relevant PROMs (23).

Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) Registry

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The PROFILES registry is a Netherlands-based registry of cancer survivors and collects information from patients will all stages of melanoma. It exists to identify outcome and psychosocial risk assessment, analyse behavioural and biological factors associated with the outcomes of cancer treatment, and evaluate psychosocial and physical care needs in cancer survivors. It collects three PROMs: the EORTC Quality of Life Questionnaire-Cancer 30 questions (QLQ-C30), Fatigue Assessment Scale (FAS) and Hospital Anxiety and Depression Scale (HADS). In addition, it collects one PREM, the EORTC QLQ-INFO26; a measure of patient satisfaction regarding the information they receive during treatment.

The Cancer Experience Registry

Finally, the Cancer Experience Registry is a US-based registry which all cancer patients are eligible to register for, regardless of cancer type or stage. For melanoma it collects PROMs through the Patient Reported Outcome Measures Information System-29 (PROMIS-29). It also has a registry-specific survey with specific patient experience items including treatment decision making, body image and financial impact of cancer. Run by the Cancer Support Community, a not-for-profit organisation, patients self-elect to register to the Cancer Experience Registry through their online website, where they are provided with an informed consent waiver, and are invited to complete the survey. This information is used primarily for research purposes.

DISCUSSION

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The four registries identified in this review varied considerably with regards to purpose of data collection and methods for data collection and dissemination. Registries specific for melanoma, namely the DMTR and Adelphi Real-World Disease-Specific Programme (Melanoma) both favoured the use of health-related quality of life (HR-QoL) PROMs: EQ-5D, FACT-G and FACT-M in their data collection. The EQ-5D is a generic questionnaire, aimed at general populations, while the FACT-G and FACT-M are both disease specific questionnaires aimed toward people undergoing cancer therapy, and melanoma-specific treatment respectively. As such, the data collected provides a comprehensive overview of participant HR-QoL, and facilitates comparisons between registries.

PROMs collected in registries of people with non-specific cancers such as PROFILES, utilised the EORTC QLQ-C30 which is a cancer-specific measure of HR-QoL. Specific co-morbidities, such as fatigue and depression, are measured by separate instruments, namely the FAS and HADS. Conversely, the Cancer Experience Registry used PROMIS-29, a generalised PROM and paired this with registry specific items relating to the experience of melanoma treatment. PROFILES was the only identified registry that used a validated PREM, the EORTC QLQ-INFO26, assessing satisfaction with information received during treatment.

Each registry applied a different method of participant recruitment and data collection, with associated advantages and disadvantages. The majority of registries favoured an online approach (DMTR, PROFILES, and Cancer Experience Registry). The Cancer Experience Registry was unique in that it functions independently of treatment centres and existing cancer registries, with all participants volunteering to join through their online website. This presents an efficient means of collecting data, and minimises missing values and

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costs associated with active participant recruitment, printing/mailing of surveys, and data entry. However, whilst the internet has a high penetrance in many countries including Australia, internet use is associated with younger age groups and a higher socio-economic status (24). As such, this method runs the risk of sample bias.

One potential solution to this is to supplement online tools with paper-based collection; a strategy employed by the PROFILES registry. Here, patients meeting the criteria of a 'cancer survivor' within the existing Eindhoven Cancer Registry, were actively recruited via mail, through their treating physician (19). This mail package included a link to a secure website for response to the survey items, and also included a paper copy which could be mailed back, should the participant not have internet access.

Similarly, the DMTR mailed eligible participants an invitation to join the registry, however data collection was conducted either through a web-based 'patient portal' system, or collected through a method at the discretion of the participating institution, before forwarding the results to the Dutch Institute for Clinical Auditing (DICA). Here, patients were contacted regularly to provide up-to-date patient reported outcome data. One issue inherent to this collection method is that whilst patients were free to complete their feedback in their own time, the timing and environment in which it was completed is variable, and thus poorly controlled. Conversely, the Adelphi Real-World Disease-Specific Programme (Melanoma) registry had participants complete their feedback form directly following consultation with their physician, therefore minimizing bias relating to the timing of collection.

The use of large-scale clinical quality registries introduces a broad set of valuable applications. For registries involving patients undergoing active treatment, the use of PROMs

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and PREMs allow for benchmarked feedback on quality of care provided by participating centres (including primary care providers), or cancer registries, as well as providing cost-effectiveness data of new treatment options to industry and health technology agencies. It can improve accountability of melanoma treatment, by providing feedback to participating centres/physicians of the health outcomes of their patients. For example, the DMTR provided fortnightly feedback on physician's quality of care (24). This can translate into a change in clinical care for patients with poor health outcomes. These registries also provide a robust platform for research and can be used in annual reports on the quality of life of people with melanoma.

A key challenge of implementing PROMs and PREMs in a melanoma registry on a national scale is their cost. For example, the costs associated with the DMTR registry are data entry, analysis, validation, and training staff in data management; estimating a cost of approximately €500 per patient registered (25). These costs might be reduced by developing strict criteria regarding which items are relevant to all registered patients, and which items are relevant only to a subset of those registered. Efficiency of data collection via integration with electronic health records and data linkage is another key consideration in reducing costs (25). As discussed previously, web-based collection provides a cost-efficient method of data acquisition, though carries with it the risk of introducing sample bias. Importantly, self-assessment tools have been shown to have similar reliability whether administered online or paper-based (26-28). As such, combining online and paper-based collection methods are a feasible compromise and may result in an increased volume of completed questionnaires, as illustrated by the PROFILES registry.

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While this review suggests registry-collection of PROMs/PREMs is feasible and describes various methods by which large-scale melanoma registries can integrate PROMs and PREMs, one limitation is the lack of evidence regarding how PROM and PREM data is currently being utilized in daily clinical practice. This is largely due to the fact that local (non-registry) datasets are more likely to be used in clinical care, and not identified by systematic review. Nonetheless, insights into the use of PROMs and PREMs in clinical practice with respect to melanoma diagnosis and treatment may provide useful strategies for data collection and feedback at a registry level.

In summary, current examples provide evidence that clinical quality registries can feasibly incorporate PROMs and PREMs for melanoma care. Specific instruments such as the EQ-5D, FACT-M and QLQC-INFO26 may be useful to integrate into the new MelCOR, although further research is required to evaluate the acceptability of these, particularly in primary care. Cost and sample bias may limit generalisability. Importantly, routine assessment and identification of patients who have poor PROMs and PREMs provides an opportunity to deliver a clinical intervention that may improve HR-QoL, improving overall melanoma care for patients.

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COMPETING INTERESTS STATEMENT

None declared.

AUTHOR CONTRIBUTIONS

Zachary Blood: Search strategy design, article screening, data extraction/analysis and manuscript preparation. Anh D Tran: Article screening, interpretation of results and reviewing the final manuscript. Lauren A Caleo: Interpretation of results and reviewing the final manuscript. Robyn PM Saw: Study design, Interpretation of results and reviewing the final manuscript. Mbathio Dieng: Study design, interpretation of results, drafting and reviewing the final manuscript. Mark Shackleton: Study design, interpretation of results and reviewing the final manuscript. H. Peter Soyer: Study design, interpretation of results and reviewing the final manuscript. Christopher Arnold: Study conception, interpretation of results and reviewing the final manuscript. Graham J Mann: Study design, interpretation of results and reviewing the final manuscript. Rachael L Morton: Study conception and design, student supervision, interpretation of results and reviewing the final manuscript.

All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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DATA STATEMENT

All data extracted from included studies is contained in Tables 1 and 2 of this manuscript.

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Table 1. Summary of included studies.

Author and Year of Study	Country	Registry featured	Study Design	Population	n (total)	PROMs/PREMs reported
Byrne et al, 2017 (10)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, breast cancer, gastric, non-small cell lung and prostate cancer patients	4899	FACT-G
Hallworth et al, 2017 (11)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, non-small cell lung and prostate cancer patients	4899	FACT-G
Rider et al, 2017 (12)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, breast cancer, gastric, non-small cell lung and prostate cancer patients	4899	FACT-G
Buchanan et al, 2017 (13)	USA	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma patients	183	FACT-M, EQ-5D
Buzaglo et al, 2017 (14)	USA	Cancer Experience Registry	Cross-sectional	Melanoma survivors	56	Registry-specific items (“financial impact of cancer”, “cost of care communication”)
Buzaglo et al, 2016 (15)	USA	Cancer Experience Registry	Abstract	N/A	N/A	N/A
Jochems et al, 2017 (16)	The Netherlands	Dutch Melanoma Treatment Registry	Summary	N/A	N/A	N/A

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Horevoorts et al, 2017 (17)	The Netherlands	PROFILES Registry	Abstract	Survivors of melanoma, basal/squamous cell carcinoma, prostate cancer, ovary cancer, endometrial cancer, thyroid cancer, Hodgkin or non-Hodgkin lymphoma, chronic lymphatic leukemia or multiple myeloma	Not reported	EORTS QLQ-C30, DS14, HADS, FAS, EORTC INFO-25, DS14
Husson et al, 2018 (18)	The Netherlands	PROFILES Registry	Abstract	Patients with a diagnosis of colon cancer, rectum cancer, melanoma, basal/squamous cell carcinoma, endometrial cancer, ovarian cancer, prostate cancer, thyroid cancer, Hodgkin or non-Hodgkin lymphoma, chronic lymphocytic leukemia or multiple myeloma	6895	EORTC QLQ-C30
Huibertse et al, 2017 (19)	The Netherlands	PROFILES Registry	Cross-sectional	Melanoma or prostate cancer survivors	767	EORTC QLQ-C30 v3.0, MSPSS, IOC Version 2.0 ('health worries subscale'), study-specific items
Mols et al, 2012 (20)	The Netherlands	PROFILES Registry	Cross-sectional	Survivors of melanoma, colorectal cancer, Hodgkin or non-Hodgkin lymphoma, or multiple myeloma	2892	Study-specific items
Van de Wal et al, 2016 (21)	The Netherlands	PROFILES Registry	Cross-sectional	Stage I or II survivors of melanoma, colorectal, endometrial, and thyroid cancers, or Hodgkin and non-Hodgkin lymphoma	2615	IOC ('health worries subscale'), SF-36, EORTC QLQ-C30, EORTC QLQ-INFO25 (three items)

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Van de Wal et al, 2014 (22)	The Netherlands	PROFILES Registry	Cross-sectional	Stage I or II survivors of melanoma, colorectal, endometrial, and thyroid cancers, or Hodgkin and non-Hodgkin lymphoma	2615	IOC ('health worries subscale')
Van de Poll-Franse et al, 2011(19)	The Netherlands	PROFILES Registry	Summary	N/A	N/A	EORCT QLQ-C30, EORTC QLQ-INFO26, FAS, HADS

PROFILES, Patient Reported Outcomes Following Initial treatment and Long term Evaluation of Survivorship; FACT-G, Functional Assessment of Cancer Therapy – General; EORTC QLQ-C30, European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire 30; EORTC QLQ-INFO25, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 25; MSPSS, Multidimensional Scale of Perceived Social Support; IOC, Impact of Cancer; SF-36, Short Form 36 Item; FACT-M, Functional Assessment of Cancer Therapy – Melanoma; EQ-5D, EuroQol-5 Dimension; EORTC QLQ-INFO26, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 26; FAS, Fatigue Assessment Scale; HADS, Hospital Anxiety and Depression Scale; DS14, The Type-D Scale.

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Table 2. Summary of registries utilising PROM and/or PREM data, including people with melanoma.

Registry name	Country	PROMs/ PREMs collected	Collection method	Collection timepoints	Inclusion criteria	Method of participant identification	Languages offered	Funding	Data Availability
Dutch Melanoma Treatment Registry	The Netherlands	EQ-5D, FACT-G, FACT-M,	Online, facilitated by clinic	Every 3 months	Stage IIIc - IV melanoma patients	Patients referred to participating melanoma centres	Dutch	Combination of pharmaceutical companies, health insurance companies and melanoma centers	Annual report, data available to contributors by application
Adelphi Real World Disease Specific Programme (Melanoma)	UK, USA, France, Germany, Italy, Spain	EQ-5D, FACT-G, FACT-M	"Patient self-completing records" completed following consultation	Once, during defined survey periods	Stage IIIb - IV melanoma patients	Sample of patients taken from participating physicians	English French German Italian Spanish	Private company	Subscription basis
PROFILES	The Netherlands	EORTC QLQ-C30, FAS, HADS, EORTC QLQ-INFO26	Online or paper-based	1-4 times/year	Cancer survivors, following primary treatment	Recruited from the Eindhoven Cancer Registry	Dutch	Comprehensive Cancer Centre South, Centre of Research on Psychology in Somatic Diseases, The Netherlands Organisation for Scientific Research	Available for non-commercial research

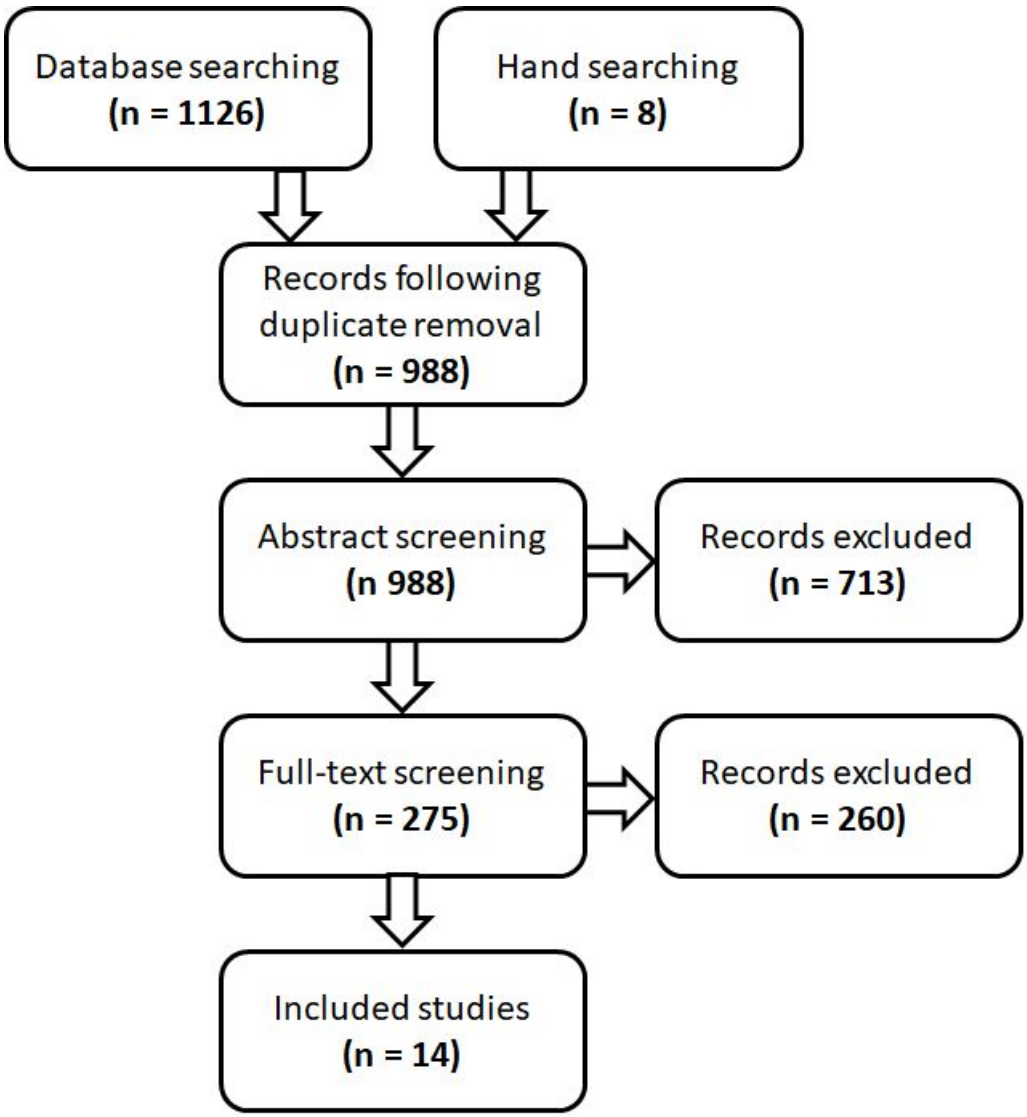
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Cancer Experience Registry	International	PROMIS-29	Online	Once, at registration	Individuals with a diagnosis of cancer, or carers of cancer patients	Self-elected participation through online website	English	Cancer Support Community	Annual report, publications from the Cancer Support Community

PROFILES, Patient Reported Outcomes Following Initial treatment and Long term Evaluation of Survivorship; FACT-G, Functional Assessment of Cancer Therapy – General; FACT-M, Functional Assessment of Cancer Therapy – Melanoma; EORTC QLQ-C30, European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire 30; FAS, Fatigue Assessment Scale; HADS, Hospital Anxiety and Depression Scale; EORTC QLQ-INFO26, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 26; PROMIS-29, Patient Reported Outcome Measures Information System-29. NB. The DS14 was not a part of the core set of PROMs included in PROFILES

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Figure 1. PRISMA flow chart.



Appendix 1. MEDLINE and PreMEDLINE Search Strategy

#	Search Statement
1	patient reported outcome measures/
2	((patient* or consumer* or participant*) adj4 (outcome* or satisf* or experienc* or assess* or symptom-measure*)).tw.
3	((patient* or consumer* or participant*) adj4 report*).tw.
4	2 and 3
5	Quality of Life/
6	(Quality-of-life or QoL or HR$QoL).tw.
7	5 or 6
8	3 and 7
9	(prom? or prem?).tw.
10	1 or 4 or 8 or 9
11	exp Melanoma/
12	(melanoma* or lentigo-maligna*).tw.
13	11 or 12
14	10 and 13

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			

Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2-3
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4
Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	6
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	7-8
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study	5-6

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		authors to identify additional studies) and date last searched.	
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	23
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	7
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	6-7
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	7, 21-22
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	7
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A narrative synthesis

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Planned methods of analysis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	7
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Figure 1
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	18-20 (Table 1)
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	N/A
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	21, (Table 2)

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Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	21, (Table 2)
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-10
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	10
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	15

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IMPLEMENTATION OF PATIENT REPORTED OUTCOME MEASURES AND PATIENT REPORTED EXPERIENCE MEASURES IN MELANOMA CLINICAL QUALITY REGISTRIES: A SYSTEMATIC REVIEW

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IMPLEMENTATION OF PATIENT REPORTED OUTCOME MEASURES AND PATIENT REPORTED EXPERIENCE MEASURES IN MELANOMA CLINICAL QUALITY REGISTRIES: A SYSTEMATIC REVIEW

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Quality of life, Clinical Audit

ABSTRACT

Objectives: To identify patient-reported outcome measures (PROMs) and patient-reported experience measures (PREMs) in clinical quality registries, for people with cutaneous melanoma, to inform a new Australian Melanoma Clinical Outcomes Registry (MelCOR); and describe opportunities and challenges of routine PROMs/PREMs collection, especially in primary care.

Design: Systematic review.

Primary and secondary outcome measures: Which PROMs and PREMs are utilised in clinical quality registries for people with cutaneous melanoma, how they are collected, frequency of collection, participant recruitment methods and funding models for each registry.

Results: 1134 studies were identified from MEDLINE, PreMEDLINE, Embase, PsychInfo, Cochrane DARE databases and TUFTS Cost-Effectiveness Analysis Registry, alongside grey literature, from database inception to 5th February 2020. Following screening, 14 studies were included, identifying four relevant registries: Dutch Melanoma Registry, Adelphi Real-World Disease-Specific Programme™ (Melanoma), Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) Registry, and Cancer Experience Registry. These utilised seven PROMs: EuroQol- 5 Dimensions (EQ-5D), Functional Assessment of Cancer-General (FACT-G) and -Melanoma (FACT-M), European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-C30 (EORTC QLQ-C30), Fatigue Assessment Scale (FAS), Hospital Anxiety and Depression Scale (HADS), Patient

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Reported Outcome Measures Information System-29 (PROMIS-29) and one PREM; EORTC Quality of Life Questionnaire – Information Module 26 (EORTC QLQ-INFO26). PROMs/PREMs in registries were reported to improve transparency of care; facilitate clinical auditing for quality assessment; enable cost-effectiveness analyses and create large-scale research platforms. Challenges included resource burden for data entry, and potential collection bias toward younger, more affluent respondents. Feedback from melanoma patients highlighted the relevance of PROMs/PREMs in assessing patient outcomes and patient experiences.

Conclusions: Clinical registries indicate PROMs/PREMs for melanoma care can be incorporated and address important gaps, however cost and collection bias may limit generalisability.

Trial registration: International Prospective Register Of Systematic Reviews Registration (PROSPERO) registration number CRD42018086737.

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ARTICLE SUMMARY

STRENGTHS AND LIMITATIONS OF THIS STUDY

- Patient reported outcome measures (PROMs) and patient reported experience measures (PREMs), have been highlighted by melanoma consumers (patients and carers) as important measures of quality that should be incorporated into clinical

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quality registries. Melanoma patients provided feedback on the importance of this review and its relevance to patient outcomes and patient experiences.

- This study captures publications of PROM and PREM collection across a range of database types.
- Provides insight into PROM and PREM collection as described by authors directly involved in their respective registry, where published.
- Only clinical quality registries in the public domain with melanoma-specific search terms were identifiable and included.
- Evaluation of the validity of identified PROMs/PREMs from primary studies was not undertaken.

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INTRODUCTION

Patient-reported outcome measures (PROMs) are described as "measurements of any aspect of a patient's health status that come directly from the patient".(1) These tools are commonly used to assess patients' self-perception of their functional status, health or quality of life. Patient-reported experience measures (PREMs) focus on patients' satisfaction with care and cover topics such as dignity and respect; consistency and coordination of care; adequate involvement in, and explanation of care; trust and communication with nurses and doctors; and satisfactory discharge information.(2) Whilst these measures were originally developed for use in research, increasingly they are being utilised for broader purposes, such as inclusion in routine healthcare or clinical quality registries for quality assurance and benchmarking.(3)

PROMs and PREMs can be applied in several stages, such as initial patient evaluations, during treatment, and in follow-up.(4) These tools can complement the existing roles of clinical quality registries as platforms for quality assessment and benchmarking, as well as for large-scale research projects. In the case of cancer, the Prostate Cancer Outcomes Registry – Australia and New Zealand (PCOR-ANZ) uses the 26-item Expanded Prostate Cancer Index Composite (EPIC-26) survey to assess the impact of diagnosis and treatment in men with prostate cancer towards improving patient wellbeing.(5) Data collected by this registry has resulted in tailored strategies and targeted programs to address important gaps in patient care.(6, 7)

With the support from key partner organisations, funding was raised through the

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Melanoma March event to develop the Australian Melanoma Clinical Outcomes Registry (MelCOR). Collection of PROMs and PREMs is an important goal of MelCOR, however challenges in this are apparent. PROMs and PREMs are not routinely collected as a part of melanoma care (including primary care) in Australia, which tends to focus mostly on traditional outcomes such as prevention of recurrence, diagnosing subsequent primary melanomas (8) and maximizing life expectancy after diagnosis. Surgical morbidity parameters such as wound complications, pain and numbness are often added to these traditional measures.(9) However, patient reports of diagnosis and treatment experiences are typically not evaluated, despite these being an important indicator of quality of care. For example, patient's self-perception of keloid scarring following diagnostic excision or other surgery on the face or hands may be associated with significant psychological morbidity including anxiety, poor self-esteem and a decreased quality of life.(10)

As the value and usefulness of MelCOR will be enhanced by inclusion of PROMs and PREMs, we sought to understand which tools capture the most relevant generic and disease-specific information for people with cutaneous melanoma, and the opportunities and challenges associated with routine PROMs and PREMs collection. The aim of this systematic review was to identify PROMs and PREMs that are used in clinical quality registries and that might be applied to patients with early-stage cutaneous melanoma, including in primary care and specialist settings where much of the initial evaluation and management of this cancer occurs.

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METHODS

A systematic review of relevant literature was conducted. The search strategy included patients treated for cutaneous melanoma of any stage, to maximise the number of potential identifiable registries in this disease context, along with articles discussing the advantages and challenges of PROMs or PREMs implementation in registries. Five databases were searched from inception to December 2017: MEDLINE (including PreMEDLINE), Embase, PsychInfo, Cochrane Database of Abstracts of Reviews of Effects (DARE) , and TUFTS Cost-Effectiveness Analysis Registry, without language restriction. Reference lists of relevant articles were hand searched for additional publications. Grey literature including digital theses, government reports (e.g. the Australian Quality and Safety Commission), and industry reports (e.g. International Consortium for Health Outcomes Measurement [ICHOM]) were also searched. A secondary search of MEDLINE (including PreMEDLINE), EMBASE, PsychInfo and the Cochrane Database was conducted from December 2017 until 3rd February 2020. Guidelines for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (11) were followed where applicable. The review was registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42018086737).

Patient and Public Involvement Statement

Two melanoma patients provided feedback on the importance of this review and its relevance to patient outcomes. The PROSPERO protocol for the systematic review was circulated to melanoma consumer representatives, then face to face feedback about PROMs/PREMs in the

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new MelCOR registry was sought. This occurred alongside the MelCOR Steering Committee meeting in 2018.

Search strategy

'Patient reported outcome measures' and 'patient reported experience measures' (including abbreviations PROMs or PREMs), along with related keywords for patients (consumers, participants) and outcomes (satisfaction, experience, assessment, symptom-measure), or quality of life, were combined with Boolean logic (AND/OR) with terms for melanoma (including lentigo maligna), to identify relevant studies (Appendix 1). Medical Subject Headings were adapted to each database as appropriate (Figure 1 – PRISMA Flow chart).

Eligibility criteria

Publications involving the use of PROMs and/or PREMs in clinical registries reporting adults with cutaneous melanoma of any disease stage (i.e. stage 0-IV), were included. PROMs and PREMs completed by proxy (e.g. caregivers) were also included. Mixed cancer populations that included cutaneous melanoma were eligible, however non-cutaneous melanoma populations (e.g. uveal melanoma), and patients with non-melanoma skin cancers only, were excluded. Studies that only used clinician-reported outcomes were excluded, as were PROMs/PREMs collected solely for research purposes (e.g. within a clinical trial). All primary study types were eligible, and grey literature was also searched, with government reports and digital theses included. Cost-effectiveness analyses were included up to the full-

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text screening stage, and their reference lists hand-searched for relevant articles. Review articles were excluded.

Study screening and data extraction

Citations and abstracts were imported into Covidence (Cochrane systematic review management program) for screening purposes. Title and abstracts were screened by two independent reviewers (ZB, AT), with discrepancies resolved through discussion, and consultation with a third reviewer (RLM) where necessary. Full-text articles were assessed if the paper was deemed to meet the inclusion criteria, was a cost-effectiveness study, or the abstract did not contain sufficient information. Clinical registries from included studies which utilised PROMs and/or PREMs data collection from people with melanoma of any stage were identified for data extraction. For each clinical registry identified through this review, the PROMs/PREMs used, patient demographics, method of collection, collection time points, and purpose of the registry were identified. Any descriptive text relating to advantages or challenges of registry collection were also extracted. The consensus-based standards for the selection of health status measurement instruments (COSMIN) four-point checklist and International Society for Quality of Life Research (ISOQOL) user's guide to implementing patient-reported outcome assessment in clinical practice were not used to assess instrument quality, as a critique of PROMs/PREMs used in registries was not an aim of this systematic review. Likewise, quality assessment of individual studies was not performed, as the role of this review was to identify relevant clinical registries present in published works, and the PROMs/PREMs they collected. Risk of bias in individual studies was described; studies were not excluded due to bias.

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RESULTS

Database and hand-searching yielded 1134 articles, with 988 articles remaining after removal of duplicates (Figure 1). Following abstract and full-text screening, 14 relevant articles remained for analysis.(12-25) These studies used either data collected from these registries to answer a research question, or were articles commenting on the advantages and challenges of PROM/PREM collection and implementation via a registry. A summary is provided in Table 1. Four relevant clinical registries were identified, each from published literature: The Dutch Melanoma Treatment Registry (DMTR), Adelphi Real-World Disease-Specific Programme (Melanoma), the Cancer Experience Registry, and the Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) registry. One registry, the Eindhoven Cancer Registry, was cited in the context of its utility for recruiting participants to the PROFILES registry.(23) This registry was not included in our analysis, however, as it did not meet our inclusion criteria of collecting PROM/PREM data from people with melanoma. Across the four included registries, ten PROMs and one PREM were used. A summary of findings including PROMs/PREMs collection method, frequency, participant recruitment and funding for each registry is presented in Table 2.

The Dutch Melanoma Treatment Registry

The Dutch Melanoma Treatment Registry is a registry of patients based in the Netherlands with unresectable stage IIIc or IV melanoma. It is used for clinical auditing, improving the transparency of quality of care in the treatment of melanoma, to detail the

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effects and costs associated with treatment, and serves as a database for melanoma research. This registry collects three PROMs: EuroQol- 5 Dimensions (EQ-5D), Functional Assessment of Cancer Therapy – General (FACT-G)(26) and Functional Assessment of Cancer Therapy -- Melanoma (FACT-M).(27) It also collects treatment activity through a Resource Use Questionnaire–Melanoma (RUQ-M). The Dutch Melanoma Treatment Registry aims to collect PROMs every three months and the RUQ-M every six months. Questions about employment and time lost from work due to melanoma and its treatment are used to quantify productivity gains and losses, and are often used in economic evaluations undertaken from a societal perspective. Data are prospectively collected using a web-based program from patients diagnosed with metastatic melanoma who attend participating treatment centres.

The Adelphi Real-World Disease Specific Programme (Melanoma)

The Adelphi Real-World Disease-Specific Programme (Melanoma) is a privately owned international healthcare database based in the UK, used for research purposes. Adelphi collates disease specific programmes on over 60 diseases, including melanoma, with data used for both medical and market research. The Melanoma programme collects three PROMs; the EQ-5D, FACT-G and FACT-M from patients with all stages of melanoma under the care of participating clinicians, from multiple countries. Here, participating clinicians are asked to collect data on the next 10 patients being consulted for the disease of interest, with these patients asked to self-complete the relevant PROMs.(28)

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Patient-Reported Outcomes Following Initial treatment and Long-term Evaluation of Survivorship (PROFILES) Registry

The PROFILES registry is a Netherlands-based registry of cancer survivors and collects information from patients will all stages of melanoma. It exists to identify outcome and psychosocial risk assessment, analyse behavioural and biological factors associated with the outcomes of cancer treatment, and evaluate psychosocial and physical care needs in cancer survivors. It collects three PROMs: the EORTC Quality of Life Questionnaire-Cancer 30 questions (QLQ-C30),(29) Fatigue Assessment Scale (FAS)(30) and Hospital Anxiety and Depression Scale (HADS).(31) In addition, it collects one PREM, the EORTC QLQ-INFO26;(32) a measure of patient satisfaction regarding the information they receive during treatment.

The Cancer Experience Registry

Finally, the Cancer Experience Registry is a US-based registry which all cancer patients are eligible to register for, regardless of cancer type or stage. For melanoma it collects PROMs through the Patient Reported Outcome Measures Information System-29 (PROMIS-29).(33) It also has a registry-specific survey with specific patient experience items including treatment decision making, body image and financial impact of cancer. Run by the Cancer Support Community, a not-for-profit organisation, patients self-elect to register to the Cancer Experience Registry through their online website, where they are provided with an informed consent waiver, and are invited to complete the survey. This information is used primarily for research purposes.

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DISCUSSION

The four registries identified in this review varied considerably with regards to purpose of data collection and methods for data collection and dissemination. Registries specific for melanoma, namely the Dutch Melanoma Treatment Registry and Adelphi Real-World Disease-Specific Programme (Melanoma) both favoured the use of health-related quality of life (HR-QoL) PROMs: EQ-5D, FACT-G and FACT-M in their data collection. The EQ-5D is a generic questionnaire, aimed at general populations, while the FACT-G and FACT-M are both disease specific questionnaires aimed toward people undergoing cancer therapy, and melanoma-specific treatment respectively. As such, the data collected provides a comprehensive overview of participant HR-QoL and facilitates comparisons between registries.

PROMs collected in registries of people with mixed cancer types such as PROFILES, utilised the EORTC QLQ-C30 which is a cancer-specific measure of HR-QoL. Specific comorbidities, such as fatigue and depression, are measured by separate instruments, namely the FAS and HADS. Conversely, the Cancer Experience Registry used PROMIS-29, a generalised PROM and paired this with registry specific items relating to the experience of melanoma treatment. PROFILES was the only identified registry that used a validated PREM, the EORTC QLQ-INFO26, assessing satisfaction with information received during treatment.

We noted registries tended to utilise PROMs developed in their geographical regions. EU-based registries utilised measures developed in the EU, such as those from the EORTC and

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EuroQol Group, whereas the US-based Cancer Experience Registry utilised the US-developed PROMIS-29. Given calibrated PROMs assessing similar constructs can be cross-walked, it is not necessary for registries to select identical measures to be considered comparable with one another. However, restricting patient-reported outcomes to validated PROMs or PREMs, as opposed to non-standardised questionnaires, is important for any findings to be generalisable.

Each registry applied a different method of participant recruitment and data collection, with associated advantages and disadvantages. The majority of registries favoured an online approach (Dutch Melanoma Treatment Registry, PROFILES, and Cancer Experience Registry). The Cancer Experience Registry was unique in that it functions independently of treatment centres and existing cancer registries, with all participants volunteering to join through their online website. This presents an efficient means of collecting data, and minimises missing values and costs associated with active participant recruitment, printing/mailing of surveys, and data entry. However, whilst the internet has a high penetrance in many countries including Australia, internet use (and smart phone apps) are associated with younger age groups and a higher socio-economic status.(34) As such, this method may run the risk of sample bias.

One potential solution to this is to supplement online tools with paper-based collection, a strategy employed by the PROFILES registry. Here, patients meeting the criteria of a 'cancer survivor' within the existing Eindhoven Cancer Registry, were actively recruited via mail, through their treating clinician.(23) This mail package included a link to a secure website for response to the survey items, and also included a paper copy which could be

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mailed back, should the participant not have internet access. Another alternative to online collection could involve a call-based interactive voice-response system, although this method was not observed in the identified registries.

Similarly, the Dutch Melanoma Treatment Registry mailed eligible participants an invitation to join the registry, however data collection was conducted either through a web-based 'patient portal' system or collected through a method at the discretion of the participating institution, before forwarding the results to the Dutch Institute for Clinical Auditing. Here, patients were contacted regularly to provide up-to-date patient reported outcome data. One issue inherent to this collection method is that whilst patients were free to complete their feedback in their own time, the timing and environment in which it was completed would have been variable, and thus poorly controlled. Conversely, the Adelphi Real-World Disease-Specific Programme (Melanoma) registry had participants complete their feedback form directly following consultation with their clinician, therefore minimizing bias relating to the timing of collection.

The use of large-scale clinical quality registries introduces a broad set of valuable applications. For registries involving patients undergoing active treatment, the use of PROMs and PREMs allow for benchmarked feedback on quality of care and medical practise variation provided by participating centres (including primary care providers), or cancer registries, as well as providing cost-effectiveness data of new treatment options to industry (e.g. medical and pharmaceutical companies) and health technology agencies such as Medical Services Advisory Committee in Australia. It can improve accountability of melanoma treatment, by providing feedback to participating centres/clinicians of the health outcomes of their patients.

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For example, the Dutch Melanoma Treatment Registry provided fortnightly feedback on clinician's quality of care.(20) This can translate into a change in clinical care for patients with poor health outcomes. These registries also provide a robust platform for research and can be used in annual reports on the quality of life of people with melanoma.

A key challenge of implementing PROMs and PREMs in a melanoma registry on a national scale is their cost. For example, the costs associated with the Dutch Melanoma Treatment Registry are data entry, analysis, validation, and training staff in data management, estimated to amount to €500 per patient registered.(35) These costs might be reduced by developing strict criteria regarding which items are relevant to all registered patients, and which items are relevant only to a subset of those registered. Efficiency of data collection via integration with electronic health records and data linkage is another key consideration in reducing costs.(35) As discussed previously, web-based collection provides a cost-efficient method of data acquisition, though carries with it the risk of introducing sample bias. Importantly, self-assessment tools have been shown to have similar reliability whether administered online or paper-based.(36, 37) As such, combining online and paper-based collection methods are a feasible compromise and may result in an increased volume of completed questionnaires, as illustrated by the PROFILES registry. With respect to funding, the registries identified here were primarily funded by private companies, non-profit organisations, or directly from melanoma treatment/research centres.

While this review suggests registry-collection of PROMs/PREMs is feasible and describes various methods by which large-scale melanoma registries can integrate PROMs and PREMs, our analysis is limited to clinical quality registries identified from published and grey literature.

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Although the study search was comprehensive and we did not apply a language restriction, we are aware that other clinical quality registries may exist (e.g. in hospital or health service databases), however any such registries outside the public domain were unable to be included. Reporting bias is also possible, if primary study authors did not use common search terms. A further limitation of our study was the inability to report on the validation of each PROM/PREM for use in melanoma care, due to incomplete reporting and assessment of instrument validation in the primary studies.

We found there was little evidence of how PROM and PREM data is currently being utilized in daily clinical practice. This is largely due to the fact that local (non-registry) datasets are more likely to be used in clinical care, and not identified by systematic review. For example the ePROMs-MEL study that is currently underway in Australia.(38) Developing future registries within electronic health records could overcome this and provide greater integration with existing online health systems. Nonetheless, insights into the use of PROMs and PREMs in clinical practice with respect to melanoma diagnosis and treatment may provide useful strategies for data collection and feedback at a registry level. Our future plan for MelCOR is to pilot the feasibility and acceptability of the above-mentioned PROMs and PREMs with the addition of a new validated instrument, Melanoma Concerns Questionnaire (MCQ-28),(39, 40) among patients, clinicians, researchers and registry stakeholders.

In summary, current examples provide evidence that clinical quality registries can feasibly incorporate PROMs and PREMs for melanoma care and will inform how best to implement these tools in our new MelCOR. Specific instruments such as the EQ-5D, FACT-M and QLQC-INFO26 may be useful to integrate into MelCOR, although further research as outlined above,

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is required to evaluate the acceptability of these, particularly in primary care. The integration of a general PROM, a disease-specific PROM and a PREM in future registries would provide a wide breadth of data for quality assurance, to minimise clinical practice variation and as a research platform. Cost and sample bias may limit generalisability. Importantly, routine assessment and identification of patients who have poor PROMs and PREMs provides an opportunity to deliver a clinical intervention that may improve HR-QoL, improving overall melanoma care for patients.

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COMPETING INTERESTS STATEMENT

RPMS has received honoraria for advisory board participation from MSD, Novartis and Qbiotics and speaking honoraria from BMS.

HPS is a shareholder of MoleMap NZ Limited and e-derm consult GmbH, and undertakes regular teledermatological reporting for both companies. HPS is a Medical Consultant for Canfield Scientific Inc., Revenio Research Oy and also a Medical Advisor for First Derm.

CA is a Director of BioGrid Australia Ltd.

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AUTHOR CONTRIBUTIONS

Zachary Blood: Search strategy design, article screening, data extraction/analysis and manuscript preparation. Anh D Tran: Article screening, interpretation of results and reviewing the final manuscript. Lauren A Caleo: Interpretation of results and reviewing the final manuscript. Robyn PM Saw: Study design, Interpretation of results and reviewing the final manuscript. Mbathio Dieng: Study design, interpretation of results, drafting and reviewing the final manuscript. Mark Shackleton: Study design, interpretation of results and reviewing the final manuscript. H. Peter Soyer: Study design, interpretation of results and reviewing the final manuscript. Christopher Arnold: Study conception, interpretation of results and reviewing the final manuscript. Graham J Mann: Study design, interpretation of results and reviewing the final manuscript. Rachael L Morton: Study conception and design, student supervision, interpretation of results and reviewing the final manuscript.

All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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DATA STATEMENT

All data extracted from included studies is contained in Tables 1 and 2 of this manuscript.

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FIGURE LEGENDS

Figure 1. PRISMA flow chart.

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Table 1. Summary of included studies.

Author and Year of Study	Country	Registry featured	Study Design	Population	n (total)	PROMs/PREMs reported
Byrne et al, 2017 (10)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, breast cancer, gastric, non-small cell lung and prostate cancer patients	4899	FACT-G
Hallworth et al, 2017 (11)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, non-small cell lung and prostate cancer patients	4899	FACT-G
Rider et al, 2017 (12)	UK	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma, breast cancer, gastric, non-small cell lung and prostate cancer patients	4899	FACT-G
Buchanan et al, 2017 (13)	USA	Adelphi Real-World Disease-Specific Programme (Melanoma)	Cross-sectional	Melanoma patients	183	FACT-M, EQ-5D
Buzaglo et al, 2017 (14)	USA	Cancer Experience Registry	Cross-sectional	Melanoma survivors	56	Registry-specific items (“financial impact of cancer”, “cost of care communication”)
Buzaglo et al, 2016 (15) (Abstract)	USA	Cancer Experience Registry	Cross-sectional	N/A	N/A	N/A
Jochems et al, 2017 (16) (Summary)	The Netherlands	Dutch Melanoma Treatment Registry	Longitudinal cohort	Melanoma patients – all stages	1472	EQ-5D-3L, FACT-G, FACT-M

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Horevoorts et al, 2017 (17) (Abstract)	The Netherlands	PROFILES Registry	Cross-sectional	Survivors of melanoma, basal/squamous cell carcinoma, prostate cancer, ovary cancer, endometrial cancer, thyroid cancer, Hodgkin or non-Hodgkin lymphoma, chronic lymphatic leukemia or multiple myeloma	Not reported	EORTS QLQ-C30, DS14, HADS, FAS, EORTC INFO-25, DS14
Husson et al, 2018 (18) (Abstract)	The Netherlands	PROFILES Registry	Cross-sectional	Patients with a diagnosis of colon cancer, rectum cancer, melanoma, basal/squamous cell carcinoma, endometrial cancer, ovarian cancer, prostate cancer, thyroid cancer, Hodgkin or non-Hodgkin lymphoma, chronic lymphocytic leukemia or multiple myeloma	6895	EORTC QLQ-C30
Huibertse et al, 2017 (19)	The Netherlands	PROFILES Registry	Cross-sectional	Melanoma or prostate cancer survivors	767	EORTC QLQ-C30 v3.0, MSPSS, IOC Version 2.0 ('health worries subscale'), study-specific items
Mols et al, 2012 (20)	The Netherlands	PROFILES Registry	Cross-sectional	Survivors of melanoma, colorectal cancer, Hodgkin or non-Hodgkin lymphoma, or multiple myeloma	2892	Study-specific items
Van de Wal et al, 2016 (21)	The Netherlands	PROFILES Registry	Cross-sectional	Stage I or II survivors of melanoma, colorectal, endometrial, and thyroid cancers, or Hodgkin and non-Hodgkin lymphoma	2615	IOC ('health worries subscale'), SF-36, EORTC QLQ-C30, EORTC QLQ-INFO25 (three items)

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Van de Wal et al, 2014 (22)	The Netherlands	PROFILES Registry	Cross-sectional	Stage I or II survivors of melanoma, colorectal, endometrial, and thyroid cancers, or Hodgkin and non-Hodgkin lymphoma	2615	IOC ('health worries subscale')
Van de Poll-Franse et al, 2011 (23) (Summary)	The Netherlands	PROFILES Registry	Cross-sectional	N/A	N/A	EORCT QLQ-C30, EORTC QLQ-INFO26, FAS, HADS

PROFILES, Patient Reported Outcomes Following Initial treatment and Long term Evaluation of Survivorship; FACT-G, Functional Assessment of Cancer Therapy – General; EORTC QLQ-C30, European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire 30; EORTC QLQINFO25, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 25; MSPSS, Multidimensional Scale of Perceived Social Support; IOC, Impact of Cancer; SF-36, Short Form 36 Item; FACT-M, Functional Assessment of Cancer Therapy – Melanoma; EQ-5D, EuroQol-5 Dimension; EORTC QLQ-INFO26, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 26; FAS, Fatigue Assessment Scale; HADS, Hospital Anxiety and Depression Scale; DS14, The Type-D Scale.

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Table 2. Summary of registries utilising PROM and/or PREM data, including people with melanoma.

Registry name	Country	PROMs/ PREMs collected	Collection method	Collection timepoints	Inclusion criteria	Method of participant identification	Languages offered	Funding	Data Availability
Dutch Melanoma Treatment Registry	The Netherlands	EQ-5D, FACT-G, FACT-M,	Online, facilitated by clinic	Every 3 months	Stage IIIc - IV melanoma patients	Patients referred to participating melanoma centres	Dutch	Combination of pharmaceutical companies, health insurance companies and melanoma centers	Annual report, data available to contributors by application
Adelphi Real World Disease Specific Programme (Melanoma)	UK, USA, France, Germany, Italy, Spain	EQ-5D, FACT-G, FACT-M	"Patient selfcompleting records" completed following consultation	Once, during defined survey periods	Stage IIIb - IV melanoma patients	Sample of patients taken from participating physicians	English French German Italian Spanish	Private company	Subscription basis
PROFILES	The Netherlands	EORTC QLQ-C30, FAS, HADS, EORTC QLQ-INFO26	Online or paper-based	1-4 times/year	Cancer survivors, following primary treatment	Recruited from the Eindhoven Cancer Registry	Dutch	Comprehensive Cancer Centre South, Centre of Research on Psychology in Somatic Diseases, The Netherlands Organisation for Scientific Research	Available for non-commercial research

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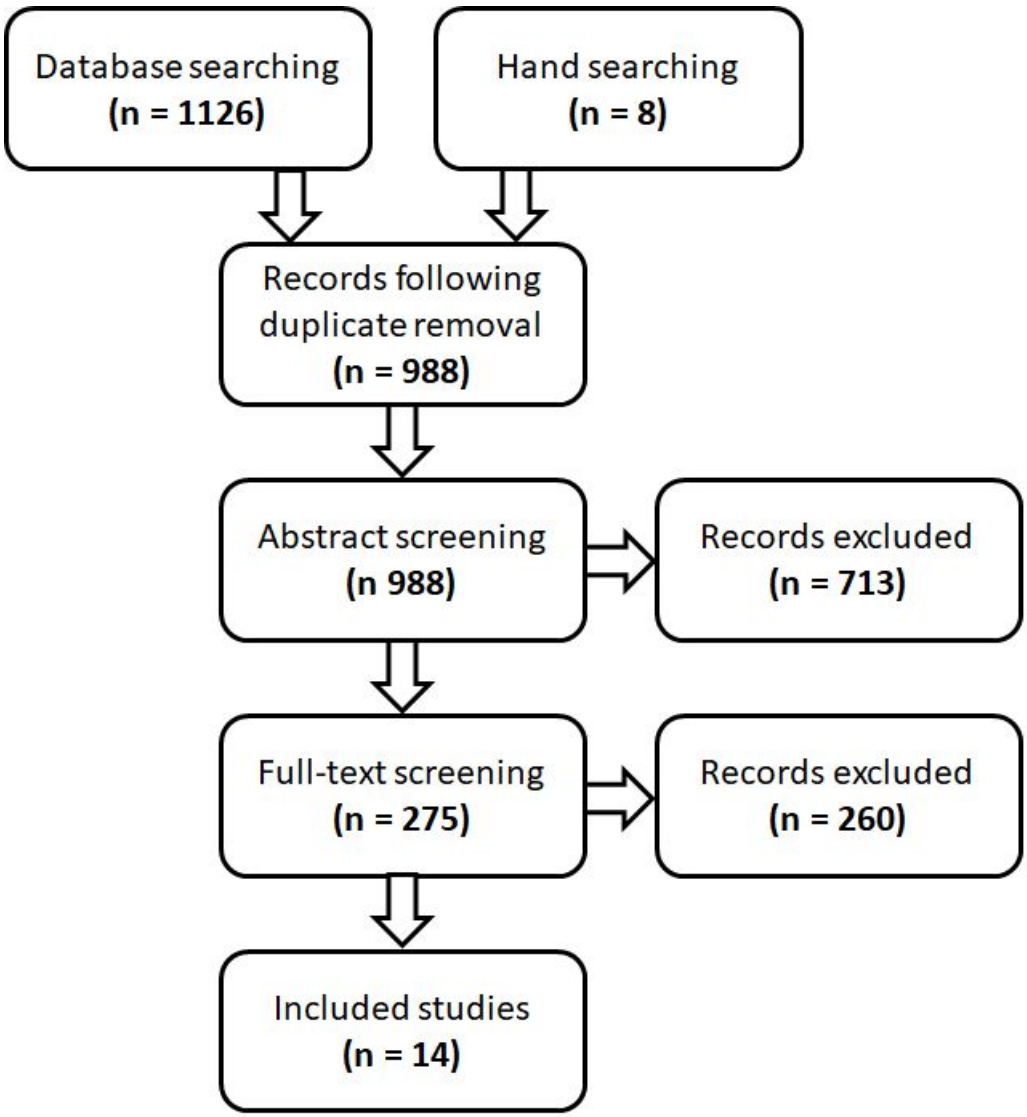
Cancer Experience Registry	International	PROMIS-29	Online	Once, at registration	Individuals with a diagnosis of cancer, or carers of cancer patients	Self-elected participation through online website	English	Cancer Support Community	Annual report, publications from the Cancer Support Community

PROFILES, Patient Reported Outcomes Following Initial treatment and Long term Evaluation of Survivorship; FACT-G, Functional Assessment of Cancer Therapy – General; FACT-M, Functional Assessment of Cancer Therapy – Melanoma; EORTC QLQ-C30, European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire 30; FAS, Fatigue Assessment Scale; HADS, Hospital Anxiety and Depression Scale; EORTC QLQ-INFO26, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire – Information Module 26; PROMIS-29, Patient Reported Outcome Measures Information System-29. NB. The DS14 was not a part of the core set of PROMs included in PROFILES

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Figure 1. PRISMA flow chart.



Appendix 1. MEDLINE and PreMEDLINE Search Strategy

#	Search Statement
1	patient reported outcome measures/
2	((patient* or consumer* or participant*) adj4 (outcome* or satisf* or experienc* or assess* or symptom-measure*)).tw.
3	((patient* or consumer* or participant*) adj4 report*).tw.
4	2 and 3
5	Quality of Life/
6	(Quality-of-life or QoL or HR$QoL).tw.
7	5 or 6
8	3 and 7
9	(prom? or prem?).tw.
10	1 or 4 or 8 or 9
11	exp Melanoma/
12	(melanoma* or lentigo-maligna*).tw.
13	11 or 12
14	10 and 13

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			

Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2-3
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4
Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	6
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	7-8
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study	5-6

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		authors to identify additional studies) and date last searched.	
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	23
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	7
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	6-7
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	7, 21-22
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	7
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A narrative synthesis

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Planned methods of analysis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	7
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Figure 1
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	18-20 (Table 1)
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	N/A
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	21, (Table 2)

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Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	21, (Table 2)
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-10
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	10
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	15

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